

Collaborative Cloud Threat Hunting: A Federated, Privacy-Preserving Security Analytics Framework

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Abstract—Enterprises operating across multiple public clouds, private data centers, and edge networks struggle to detect coordinated cyber attacks when security telemetry cannot be centralized for compliance and privacy reasons. This paper proposes a federated cyber threat hunting framework that lets distributed tenants collaboratively train detection models and correlate incidents without sharing raw logs. The architecture combines local anomaly detection, secure model aggregation, and privacy-preserving techniques such as federated learning, encrypted update exchange, and differential privacy to build cross-cloud threat intelligence. Experimental evaluation in a simulated multi-cloud setting shows that this approach improves detection of distributed attack campaigns, reduces alerting latency, and preserves regulatory data-governance constraints compared to traditional centralized security information and event management workflows.

Index Terms—Federated learning, cyber threat hunting, distributed cloud security, privacy-preserving AI, incident correlation, secure aggregation.

I. INTRODUCTION

A. The Visibility Gap in Multi-Cloud Security

Modern enterprise infrastructure rarely resembles the single-provider, single-region deployments of a decade ago. Organizations now distribute workloads across Amazon Web Services, Microsoft Azure, Google Cloud Platform, private OpenStack clusters, and thousands of edge locations. This distribution offers undeniable benefits in latency, availability, and vendor independence. Yet it fragments security telemetry across administrative boundaries, cloud providers, and geographic regions subject to conflicting data sovereignty laws.

Security operations centers face an impossible mandate: detect attacks that traverse these boundaries without ever consolidating the evidence required to see them. A credential stuffing campaign may originate from a compromised edge device in Singapore, pivot through a container workload in Frankfurt, and exfiltrate data from object storage in Virginia. Each cloud provider sees only its segment of the attack chain. Each tenant retains only its own logs. The complete picture exists nowhere.

Traditional security information and event management systems assume centralization. They ingest, normalize, and analyze logs in a single data lake. This model collapses under the constraints of modern privacy regulation. The General Data Protection Regulation restricts cross-border data transfer. The Health Insurance Portability and Accountability Act imposes strict controls on protected health information. Payment Card Industry Data Security Standard requires segmentation of card-

holder data. Enterprises cannot simply copy security telemetry across regions without violating these obligations.

B. The Promise of Federated Analytics

Federated learning emerged from a simple observation: a machine learning model requires far less information than its training data. The weights and gradients of a neural network encode statistical patterns, not raw observations. If many parties could collaboratively train a shared model while exchanging only these parameter updates, they might achieve collective intelligence without collective exposure.

This insight has transformed mobile machine learning, enabling predictive keyboards and voice assistants that learn from user interactions without uploading conversation transcripts. Its application to cybersecurity remains nascent, yet the parallel is compelling. Security telemetry is sensitive, distributed, and information-rich. The patterns that distinguish benign behavior from malicious activity—unusual outbound connections, authentication anomalies, privilege escalations—are precisely the statistical regularities that neural networks excel at extracting.

C. Research Contributions

This paper presents a complete framework for federated cyber threat hunting in multi-cloud environments. Our contributions are fourfold.

First, we propose a reference architecture that decomposes global threat detection into local sensing, federated training, and secure aggregation components. The architecture explicitly addresses the operational constraints of enterprise cloud deployments: intermittent connectivity, heterogeneous telemetry formats, and strict data residency requirements.

Second, we implement privacy-preserving mechanisms including differential privacy noise injection and secure multi-party computation for model aggregation. These mechanisms provide formal privacy guarantees measurable through epsilon-delta bounds and information leakage metrics.

Third, we evaluate the framework through simulation representing a twelve-month attack campaign across three cloud providers and forty-two geographic regions. Our experiments measure detection accuracy, communication overhead, and privacy leakage under varied federation conditions.

Fourth, we characterize the operational trade-offs inherent in federated threat hunting: convergence speed versus privacy loss, local update frequency versus bandwidth consumption, and model granularity versus statistical power.

D. Paper Organization

Section II surveys related work in distributed cloud security, federated learning, and privacy-preserving computation. Section III presents the system architecture and component specifications. Section IV describes the experimental methodology and simulation environment. Section V reports quantitative results on detection performance, privacy guarantees, and scalability. Section VI examines operational challenges including adversarial resilience and heterogeneous participation. Section VII concludes with recommendations for enterprise adoption and directions for future research.

II. RELATED WORK

A. Distributed Cloud Security Operations

The security community has long recognized the limitations of perimeter-based defense in distributed environments. Early work on cloud security emphasized identity federation and API access control, assuming that workloads remained within provider boundaries. Contemporary research acknowledges that attacks now propagate across providers, regions, and trust domains. Enterprise security monitoring typically relies on security information and event management platforms that aggregate logs into centralized data warehouses. This approach becomes infeasible at cloud scale. A single Amazon Web Services organization may generate petabytes of CloudTrail, VPC flow log, and GuardDuty finding data daily. Transporting, storing, and querying this volume across regions exceeds reasonable bandwidth and cost budgets. More fundamentally, cloud providers impose egress fees that make cross-region log transfer economically prohibitive.

Regulatory constraints further complicate centralization. The Schrems II decision invalidated the EU-US Privacy Shield framework, restricting lawful transfer of European personal data to jurisdictions without equivalent privacy protections. Enterprises operating in regulated industries—healthcare, financial services, government contracting—face additional sector-specific data localization mandates. Security operations must function within these constraints, not despite them. Mohammed examined the application of artificial intelligence for detecting and preventing financial crimes in digital payment ecosystems, demonstrating that distributed fraud detection systems must operate within strict data residency boundaries while maintaining detection efficacy [1]. This parallel from financial crime detection reinforces the fundamental challenge addressed by our federated threat hunting framework: effective security monitoring must operate across organizational boundaries without consolidating sensitive data.

Thallam conducted comparative analysis of public cloud providers for big data analytics, documenting the substantial variation in telemetry formats, API semantics, and data access controls across AWS, Azure, and Google Cloud [2]. This heterogeneity complicates cross-cloud security monitoring because normalized views require mapping disparate data models into common representations. Our framework addresses this through provider-specific feature extractors that map disparate

log formats into a common embedding space while maintaining data locality.

B. Federated Learning Foundations

Federated learning was formalized by McMahan and colleagues at Google in 2016, initially applied to next-word prediction in Gboard [8]. The canonical federated averaging algorithm proceeds in rounds: participating nodes receive the current global model, perform several epochs of local training on private data, and transmit encrypted gradient updates to a central aggregator. The aggregator computes a weighted average of received updates, applies the result to the global model, and distributes the improved model for subsequent rounds.

Several variations address specific deployment constraints. Cross-silo federated learning assumes stable, always-available participants such as hospitals or financial institutions. Cross-device federated learning accommodates millions of intermittently connected mobile devices with heterogeneous compute capabilities and battery constraints. Our framework adapts cross-silo semantics for enterprise cloud environments where tenants maintain persistent presence but experience variable workload intensities. Kairouz and colleagues provided a comprehensive survey of advances and open problems in federated learning, establishing the theoretical foundations for privacy-preserving distributed optimization [12]. Their work catalogs the trade-offs between communication efficiency, privacy guarantees, and model accuracy that inform our architectural decisions.

Li and colleagues examined challenges, methods, and future directions for federated learning, identifying heterogeneous data distributions, unreliable participant availability, and malicious participant resilience as core research problems [13]. These challenges manifest acutely in security applications where tenant workloads exhibit substantial variation and adversaries may compromise participating nodes. Yang and colleagues provided an accessible introduction to federated machine learning concepts and applications, emphasizing the distinction between horizontal federation where participants share feature spaces but differ in samples, and vertical federation where participants share samples but differ in features [14]. Our threat hunting framework implements horizontal federation across cloud tenants with common feature representations derived from normalized security telemetry.

C. Privacy-Preserving Computation in Security

Three families of privacy-preserving techniques are relevant to federated threat hunting. Secure multi-party computation enables multiple parties to jointly compute a function without revealing individual inputs. In our context, the aggregator and participating clouds jointly compute the averaged model update without exposing per-tenant gradients. Bonawitz and colleagues developed practical secure aggregation protocols for privacy-preserving machine learning, demonstrating that multi-party computation can scale to thousands of participants with acceptable communication overhead [9].

Homomorphic encryption allows computation directly on ciphertext; encrypted model updates can be aggregated without decryption. Chen and colleagues surveyed the intersection of federated learning with privacy-preserving computation, cataloging the performance characteristics and security guarantees of homomorphic encryption, secure multi-party computation, and trusted execution environments. Their analysis documents the substantial computational overhead of homomorphic operations, which informed our decision to make privacy mechanism selection configurable based on tenant compliance requirements.

Differential privacy guarantees that the inclusion or exclusion of any single record does not significantly affect model outputs, quantified by parameters epsilon and delta. Abadi and colleagues introduced differentially private stochastic gradient descent, demonstrating that carefully calibrated noise addition during training provides formal privacy guarantees with acceptable accuracy degradation [10]. Our implementation adapts their approach to the federated setting, with tenants applying differential privacy to gradient updates before transmission.

Gosselin and colleagues conducted a comprehensive survey of privacy and security in federated learning, analyzing threats including gradient leakage, membership inference, and model inversion [6]. Their work establishes that even aggregated model updates may leak information about individual training examples, motivating the privacy-preserving mechanisms incorporated in our framework.

D. Federated Learning for Cybersecurity Applications

The application of federated learning to cybersecurity remains nascent but rapidly developing. Jahromi and colleagues proposed an ensemble deep federated learning cyber-threat hunting model for Industrial Internet of Things environments, demonstrating that distributed threat detection across industrial control systems benefits from collaborative model training while preserving operational data confidentiality [3]. Their work focuses on homogeneous industrial environments rather than the heterogeneous multi-cloud deployments addressed by our framework.

Mohammed examined agentic AI as a proactive cybercrime sentinel for detecting and deterring social engineering attacks, highlighting the importance of distributed intelligence that operates at the edge while maintaining centralized coordination [5]. This architectural pattern parallels our federated approach, where local detection agents operate autonomously while contributing to globally coordinated threat intelligence.

Niu and colleagues analyzed the evolution of threats in the global risk network, demonstrating that attacks increasingly propagate across organizational and geographic boundaries [7]. Their network analysis reveals that effective threat detection requires visibility across these boundaries, precisely the capability our federated framework enables without compromising data sovereignty.

E. Adversarial Resilience and Byzantine Tolerance

Federated learning systems are vulnerable to poisoning attacks where malicious participants contribute crafted gradients to corrupt the global model. In the threat hunting context, a compromised tenant could submit updates designed to suppress detection of specific attack patterns or to create false positives that degrade operator trust. Blanchard and colleagues developed Byzantine-tolerant gradient descent algorithms that maintain convergence even when a fraction of participants behave adversarially [16]. Their coordinate-wise median and trimmed mean aggregation techniques inform our defensive mechanisms against gradient manipulation attacks.

Bagdasaryan and colleagues demonstrated practical backdoor attacks against federated learning systems, showing that adversaries controlling a small fraction of participants can implant targeted misclassifications while maintaining normal performance on benign inputs. Our simulations reproduce these findings in the threat hunting context and evaluate defensive aggregation techniques. Sun and colleagues further analyzed the feasibility of backdoor attacks in federated learning, identifying conditions under which attacks succeed or fail [20].

F. Communication Efficiency and Scalability

Communication efficiency is critical for federated learning at scale. Shamir and colleagues developed communication-efficient distributed optimization algorithms using approximate Newton-type methods, demonstrating that higher-order information reduces communication rounds. While our current implementation uses first-order methods, these advances suggest directions for future optimization.

Wang and colleagues studied adaptive federated learning in resource-constrained edge computing systems, developing protocols that adjust local update frequency based on device capabilities and network conditions [18]. Their insights inform our asynchronous federation design that accommodates intermittent tenant availability and variable workload intensities.

Zhao and colleagues analyzed the challenges of federated learning with non-IID data distributions, demonstrating that heterogeneous data across participants slows convergence and degrades final model accuracy [19]. This challenge is particularly acute in threat hunting, where tenant workloads exhibit substantial variation across industries, regions, and compliance regimes. Our framework addresses non-IID data through adaptive learning rates and stability-weighted aggregation.

III. EXPERIMENTAL METHODOLOGY AND SIMULATION ENVIRONMENT

A. Simulation Infrastructure

Evaluating federated threat hunting at enterprise scale requires infrastructure beyond typical academic computing resources. We constructed a discrete-event simulator modeling forty-two geographic regions across three cloud providers, each region hosting between five and two hundred tenant workloads. The simulation spans twelve months of operation with one-minute time granularity.

Each simulated tenant generates telemetry according to empirically observed distributions. Normal traffic patterns are derived from publicly released cloud workload traces. Attack traffic is injected according to a stochastic process calibrated to Mandiant incident response data. Attack types include credential stuffing, privileged account abuse, data exfiltration, and ransomware deployment.

The simulator implements the complete federated training protocol with configurable participation rates, network latency distributions, and privacy mechanism parameters. We executed 1,400 simulation runs requiring approximately 7,200 core-hours on a 48-node compute cluster. Results reported in Section V represent averages across ten runs with distinct random seeds.

B. Baseline Comparisons

We compare the federated framework against three baseline architectures. Centralized SIEM represents traditional security operations: all telemetry transmitted to a single data lake for analysis by a global anomaly detection model. This baseline is privacy-violating and bandwidth-prohibitive, included only to establish an upper bound on detection performance.

Isolated detection represents the current reality of multi-cloud security: each tenant operates independently with no cross-cloud intelligence sharing. Tenants train identical autoencoder architectures exclusively on local data.

Federated without privacy preserves the collaborative training structure but omits differential privacy, secure aggregation, and encryption. This baseline isolates the accuracy impact of privacy-preserving mechanisms.

C. Evaluation Metrics

Detection performance is measured through precision, recall, and F1-score at the tenant and campaign levels. Tenant-level metrics assess whether the local model correctly flags individual attack events. Campaign-level metrics assess whether the global correlator identifies multi-tenant attack campaigns as coherent incidents.

Privacy leakage is quantified through membership inference attacks. We train shadow models to distinguish whether a specific tenant contributed to a given global model round. Success rates above random guessing indicate information leakage. Differential privacy parameters epsilon and delta are reported where applicable.

Operational costs include communication bandwidth, training duration, and aggregator computational load. These are measured per round and accumulated over the simulation period.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Detection Accuracy and Attack Coverage

The federated framework substantially outperforms isolated detection across all attack types and tenant sizes. For credential stuffing campaigns spanning more than five tenants, federated F1-score averages 0.83 against 0.41 for isolated detection. The improvement stems from two mechanisms. First, the

global model learns feature representations that generalize across cloud environments; isolated models overfit to provider-specific noise patterns. Second, the correlator identifies temporally correlated anomalies that individual tenants cannot distinguish from routine variability.

Centralized SIEM achieves marginally higher F1-score than federated learning, 0.89 versus 0.83, but requires transmission of all telemetry to a single location. In privacy-constrained scenarios where centralization is prohibited, federated learning provides 92% of the detection benefit with zero data transfer.

Fig. 1: F1-score as function of participating tenant count. Federated learning approaches centralized performance while preserving data locality. Isolated detection shows minimal improvement with additional tenants.

Detection latency decreases substantially under federation. Centralized systems must wait for telemetry transmission across continents; our simulations show median detection latency of 187 seconds for cross-region attacks under centralization. Federated detection achieves 43 seconds median latency because local models flag attacks immediately upon observing anomalous behavior. Global correlation adds bounded delay for round completion, typically 60-120 seconds.

B. Privacy Guarantees and Utility Trade-offs

Differential privacy implementation incurs measurable accuracy costs. With epsilon = 2.0, a typical privacy budget for sensitive applications, F1-score decreases by 7.2% relative to non-private federation. The degradation concentrates in low-volume tenants whose contributions are overwhelmed by noise. Tenants generating fewer than 10,000 events per round experience 15-20% F1 reduction; high-volume tenants experience 3-5% reduction.

Secure multi-party computation and homomorphic encryption introduce no statistical accuracy degradation—the computations are algebraically equivalent to plaintext aggregation. However, these mechanisms impose substantial computational overhead. Homomorphic aggregation requires 340 seconds per round for 100 participants, compared to 6.8 seconds for plaintext aggregation. Secure multi-party computation completes in 47 seconds but requires six communication rounds between aggregator and participants.

Membership inference attacks achieve 52.3% accuracy against non-private federated models, marginally above random guessing baseline of 50%. This suggests that even without explicit privacy mechanisms, the aggregation of many participants provides substantial confidentiality. Differential privacy at epsilon = 2.0 reduces inference accuracy to 50.1%, statistically indistinguishable from random.

C. Communication Efficiency and Scalability

The federated framework exhibits excellent communication efficiency relative to centralized alternatives. Each participant transmits 1.2 MB per round, independent of local telemetry volume. Centralized transmission for a medium-size tenant

generating 500 GB daily would require approximately 58 MB per second sustained, exceeding typical cloud egress capacity.

Communication costs scale linearly with participant count at the aggregator and remain constant per participant. At 1,000 participants, aggregator inbound bandwidth reaches 120 MB/s, sustainable with moderate provisioning. Larger deployments would require hierarchical aggregation to avoid aggregator saturation.

Participant dropout rates up to 30% have minimal impact on model convergence. The aggregator weights remaining participants proportionally and distributes the updated model to all tenants, including those absent from training. Dropouts exceeding 50% delay convergence by approximately three rounds but do not prevent eventual convergence.

D. Attack Detection by Campaign Type

Performance varies substantially across attack categories. The framework excels at detecting credential-based attacks: brute force, password spraying, and stolen credential reuse. These attacks generate distinctive authentication failure patterns that autoencoders readily distinguish from benign variability. F1-score for credential attacks reaches 0.91.

Detection of data exfiltration is more challenging. Attackers who maintain low transfer volumes beneath per-tenant thresholds evade reconstruction error detection. Cross-tenant correlation improves detection when the same destination IP receives data from multiple sources, but sophisticated attackers rotate infrastructure. F1-score for exfiltration reaches 0.67.

Ransomware deployment detection achieves 0.74 F1-score. File encryption operations generate distinctive file modification patterns that local models recognize. Cross-tenant correlation provides less incremental benefit because ransomware campaigns typically target tenants sequentially rather than simultaneously.

TABLE I: Detection Performance by Attack Type

Attack Type	Precision	Recall	F1	Latency (s)
Credential stuffing	0.89	0.93	0.91	38
Privilege escalation	0.82	0.78	0.80	52
Data exfiltration	0.71	0.63	0.67	97
Ransomware	0.76	0.72	0.74	64
Cross-tenant scanning	0.84	0.81	0.82	29
Insider threat	0.58	0.52	0.55	124

V. OPERATIONAL CHALLENGES AND DEPLOYMENT CONSIDERATIONS

A. Adversarial Resilience and Model Poisoning

Federated learning systems are vulnerable to poisoning attacks where malicious participants contribute crafted gradients to corrupt the global model. In the threat hunting context, a compromised tenant could submit updates designed to suppress detection of specific attack patterns or to create false positives that degrade operator trust.

Our simulations include two poisoning attack variants. Gradient manipulation attacks modify local updates to reverse the direction of specific feature weights. Backdoor attacks insert

hidden triggers that cause the global model to misclassify attacker-controlled inputs while maintaining normal performance on benign traffic.

The framework demonstrates moderate resilience to gradient manipulation when participant count exceeds fifty. An adversary controlling fewer than 10% of participants shifts model weights by less than 0.5% per round, insufficient for sustained degradation. However, backdoor attacks remain challenging; 5% malicious participants successfully implant targeted misclassification with 73% success rate.

Defenses require robust aggregation techniques beyond simple averaging. We implement coordinate-wise median aggregation and trimmed mean aggregation, which discard extreme gradient values before averaging. These techniques reduce backdoor success rate to 31% with 10% malicious participants but increase convergence time by 40-60%.

B. Heterogeneous Telemetry and Concept Drift

Enterprise cloud environments evolve continuously. Workloads migrate across regions, applications update, and user behavior shifts with business cycles. These changes manifest as concept drift: the statistical distribution of telemetry shifts, rendering previously trained models obsolete.

Local models adapt to drift through continuous training on recent data. The global model faces a more difficult problem because drift occurs asynchronously across tenants. A feature that becomes predictive in one tenant's environment may remain unchanged in others.

Our implementation addresses drift through adaptive learning rates. Tenants experiencing high reconstruction error on recently validated benign data increase local learning rates to accelerate adaptation. The global aggregator weights tenant contributions by their reconstruction error stability, reducing influence of tenants undergoing rapid change.

Despite these adaptations, drift reduces sustained detection accuracy by 8-12% relative to static environments. Organizations should anticipate periodic global model retraining on freshly certified clean data, typically quarterly.

C. Regulatory Compliance and Auditability

Privacy-preserving mechanisms complicate compliance auditing. Regulators accustomed to examining raw logs for forensic investigation must adapt to systems where logs never leave tenant premises. Our framework generates cryptographic attestations for each training round: a signed statement from each participant confirming the privacy mechanisms applied and the absence of raw data transmission.

These attestations provide evidentiary basis for compliance claims but do not satisfy all regulatory requirements. GDPR Article 22 grants data subjects the right to human review of automated decisions; federated threat hunting provides no such review mechanism. Organizations must evaluate whether fully automated detection and response aligns with their regulatory obligations.

VI. CONCLUSION AND FUTURE DIRECTIONS

A. Summary of Contributions

This paper has presented a complete framework for collaborative, privacy-preserving threat hunting across distributed cloud environments. The architecture enables enterprises to detect coordinated cyber attacks that span cloud providers and geographic regions while maintaining strict data localization and compliance with privacy regulations.

Our experimental evaluation demonstrates that federated threat hunting achieves detection accuracy approaching centralized systems without transmitting raw telemetry. The framework scales to hundreds of participating tenants with manageable communication and computation costs. Differential privacy provides formal protection against membership inference at acceptable accuracy cost. Secure aggregation and homomorphic encryption offer stronger privacy guarantees with greater computational overhead.

The framework addresses a critical operational gap in contemporary cloud security. Organizations cannot centralize security telemetry across regulatory boundaries, yet attackers face no such constraints. Federated learning provides a mathematically grounded mechanism for collective defense without collective exposure.

B. Practical Recommendations

Enterprise security teams considering federated threat hunting should begin with a limited deployment across trusted environments within a single regulatory domain. Initial participants should be high-volume tenants with mature security operations that can certify clean training data. Privacy mechanisms should be introduced gradually, beginning with secure aggregation and adding differential privacy as compliance requirements dictate.

Organizations should anticipate the operational overhead of federated learning. Model retraining, hyperparameter tuning, and incident investigation require skills distinct from traditional security analytics. We recommend designating a federated learning operations team responsible for model lifecycle management.

C. Future Research Directions

Several extensions to this work merit investigation. Decentralized federation eliminating the central aggregator would further reduce trust assumptions but introduces substantial coordination challenges. Blockchain-based consensus mechanisms offer one possible approach.

Integration of structured threat intelligence into federated models could accelerate detection of emerging attack patterns. Threat intelligence providers currently distribute indicators of compromise and attack signatures; these could inform local model initialization or global correlation rules.

Federated learning for security response remains unexplored. If detection occurs collaboratively, could response also occur collaboratively? Distributed containment actions—revoking compromised credentials, isolating affected

workloads—require cross-provider coordination that current architectures lack.

Longitudinal studies of federated threat hunting in production environments will be essential to validate simulation findings. We are collaborating with three enterprises to deploy pilot implementations; results will inform subsequent framework iterations.

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